

Exploring Gender and Racial Bias in Meta’s FACET Dataset

Abstract

This study investigates gender and racial bias in Meta’s FACET dataset, a widely used benchmark in computer vision fairness research. Our project aims to examine whether annotation-level imbalances propagate into model-level performance disparities. We perform exploratory data analysis (EDA) of demographic attributes backed by significance tests to identify significant differences in representation across gender, skin tone, and age categories. To evaluate fairness mitigation, we will be applying the Visual Bias Mitigator (VB-Mitigator) as a benchmark to assess disparities in demographic parity and equal opportunity, both before and after balancing the data ourselves using techniques like SMOTE and Subgroup SMOTE. Preliminary results suggest measurable imbalances in FACET’s demographic composition and evidence that these differences may contribute to unequal performance outcomes across subgroups. This work contributes to ongoing debates about how fairness can be operationalized through measurement, data augmentation, and contextual interpretation, bridging annotation-level analysis with model-level fairness evaluation.

1. Introduction and Background

1.1 Introduction

Our project investigates potential bias in Meta’s FACET dataset, a widely used benchmark in computer vision. We will conduct exploratory data analysis (EDA) on the dataset annotations to examine whether class labels, correlations, or distributions reflect biased representations of protected classes. Because annotation choices often encode social assumptions, analyzing these label categories provides a foundation for identifying systemic patterns of bias. To validate our findings, we will use the VB-Mitigator, a tool specifically designed to measure and mitigate bias in computer vision tasks. By cross-comparing our annotation analysis with results generated by VB-Mitigator, we will evaluate whether observed disparities align with model-level performance gaps or are mitigated through bias-reduction strategies. Finally, our analysis contributes to emerging discussions about how bias manifests in multimodal machine learning systems, where visual and textual signals jointly shape model decisions. Because computer vision models can encode demographic cues even in ostensibly task-irrelevant features, understanding whether FACET’s annotation choices introduce latent correlations becomes critical. This project therefore examines not only explicit label distributions but also the implicit representational assumptions that may propagate into downstream vision models.

The significance of this project lies in bridging dataset-level analysis with mitigation frameworks, contributing to a deeper understanding of how annotated labels influence fairness in computer vision systems. As Jacobs and Wallach argue, fairness debates often hinge on how abstract constructs like “bias” and “fairness” are operationalized, and mismatches between conceptual and operational definitions can lead directly to harm [4]. By situating FACET within this measurement modeling perspective, our analysis helps reveal where annotation practices risk embedding contested constructs of gender or race.

1.2 Background

Machine vision systems increasingly shape decisions in hiring, law enforcement, advertising, and social media. However, these systems are vulnerable to replicating and amplifying societal biases. Datasets are a central point of concern: if the categories used to annotate images embed assumptions about gender, race, or identity, these choices can cascade bias into model performance. FACET relies on categorical demographic labels, such as gender presentation and skin tone, that necessarily simplify complex social identities. Tools like the Monk Skin Tone Scale [6] are designed to standardize representation, yet such standardization risks flattening nuance. As Selbst et al. note, fairness analyses often fall into an ‘abstraction trap’ when they treat these categories as neutral technical inputs rather than socially constructed descriptors [2]. Our project responds to this critique by examining how these annotation choices shape both the representational landscape of the dataset and the model behaviors built on top of it.

Previous research has shown that even datasets intended for fairness testing may reproduce labeling practices that align with binary gender norms or simplified racial categories, leading to harms in representation and classification accuracy. Suresh and Guttag’s framework illustrates how harms may arise throughout the ML lifecycle and cautions that addressing bias at only one stage risks overlooking systemic issues [3]. Tools such as VB-Mitigator were created to help practitioners analyze these issues and experiment with mitigation strategies. Our project combines annotation-level exploratory analysis with VB-Mitigator’s bias measurement framework to examine how bias manifests across both data and model stages.

2. Objectives and Research Questions

The objectives of this project are to conduct EDA of FACET annotations to identify correlations between gender, race, and dataset labels, and to use VB-Mitigator to evaluate how bias manifests in model performance while exploring mitigation strategies. By comparing annotation-level insights with model-level mitigation outcomes, the project seeks to understand how bias at the data collection stage propagates into machine learning models. Ultimately, our deliverables will include (1) annotated bias visualizations, (2) a comparative report on model fairness before and after VB-Mitigator intervention, and (3) a set of recommendations for dataset curation and model development.

This work is guided by the following research questions:

1. Do the annotation categories in the FACET dataset encode gendered or racial bias?
2. Do tests quantify annotation-level bias within the FACET dataset using descriptive statistics and significance tests?
3. Do mitigation techniques such as SMOTE and Subgroup SMOTE explore whether synthetic resampling improves fairness metrics?
4. Are there measurable disparities in model performance across demographic subgroups when using VB-Mitigator?
5. Are there fairness disparities at the model level using VB-Mitigator metrics before and after bias mitigation?
6. How well do VB-Mitigator’s mitigation techniques align with or contradict biases observed at the annotation level?

These questions reflect broader concerns in fairness research. Ashktorab et al. highlight that practitioners often differ in their interpretation of individual vs. group fairness, revealing gaps between formal fairness metrics and applied decision-making. Blandin and Kash propose a utility-based framework to generalize fairness across domains, emphasizing that fairness should be measured in terms of outcomes and counterfactuals, not just predictions [5]. Our analysis aims to contribute to this dialogue by situating FACET and VB-Mitigator within both applied and theoretical understandings of fairness.

3. Methodology

3.1 Dataset and Preprocessing

The FACET dataset provides annotations including gender presentation, skin tone (using categories aligned with the Monk Skin Tone scale), and other demographic variables. The dataset has been preprocessed by META, reducing most features of an image with a defined “class” to a set of one-hot encodings and other descriptions. Categorical variables have been encoded for analysis, and META has seemingly removed all missing or ambiguous labels. While FACET’s preprocessing choices such as the removal of ambiguous labels raise important questions about data validity, a full audit of these decisions is beyond the scope of this paper. Instead, our analysis focuses on the dataset as released, examining how the available annotations shape demographic representation and how these patterns influence downstream fairness metrics. This leaves us to examine the effects of META’s preprocessing to determine whether or not it

introduced bias into the data, as well as how that could potentially be perpetuated by model training.

3.2 Exploratory Data Analysis (EDA)

To begin our investigation, we performed Exploratory Data Analysis (EDA) on FACET to learn more about the data after preprocessing. Our methods involve visualizing distributions across demographic variables and label categories. Our EDA aims to characterize the demographic landscape of the dataset by comparing distributions of gender presentation, skin tone, and age across FACET’s label categories. These visualizations establish a baseline representation profile, enabling us to identify disproportionate subgroup frequencies and potential annotation-level biases that warrant statistical testing.

3.3 Statistical Significance Testing

To reinforce EDA findings, we perform statistical tests:

- Chi-square tests of independence to determine whether representation imbalances across gender or skin tone are statistically significant.
- Chi-Square Goodness-of-Fit Tests to show if all the protected classes are equally represented in the dataset.
- Pairwise Proportion z-Tests to show which specific pairs of categories have statistically different levels of representation.

3.5 Adversarial Debiasing and Disparate Impact Mitigation

To perform our own bias mitigation, we’ll employ Adversarial Debiasing as an in-processing mitigation technique. Adversarial debiasing introduces an adversarial network that attempts to predict sensitive attributes (e.g., gender presentation, skin tone) from model representations while the primary predictor learns to perform the main classification task. The training process is formulated as a minimax optimization problem, where the predictor minimizes its prediction loss and the adversary maximizes its ability to infer sensitive information. This dual optimization encourages the learned representations to retain task-relevant features while discarding signals correlated with protected characteristics, resulting in intrinsically fairer models [7].

This approach aligns with implicit fairness strategies described by Wan et al, where fairness is achieved through adversarial learning that removes sensitive information from the latent representation rather than rebalancing the data distribution. By directly constraining representational bias during training, adversarial debiasing mitigates bias amplification and

enhances generalization fairness even when trained on imperfect datasets [7]. Complementary to this, we apply Disparate Impact analysis as a fairness measurement to quantify group-level disparities in model outcomes. Disparate Impact assesses whether decision rates differ between protected and unprotected groups, using the ratio of favorable outcomes as an indicator of statistical parity. Ratios below the “four-fifths rule” (0.8) suggest disproportionate treatment of a subgroup [7]. By evaluating Disparate Impact before and after adversarial debiasing, we measure the reduction in group-level bias achieved by in-processing fairness optimization.

Integrating both techniques allows us to have a multi-perspective fairness assessment: adversarial debiasing serves as an in-processing corrective mechanism, while Disparate Impact provides a measurable standard for equity in outcomes. These methods, implemented alongside the VB-Mitigator framework, enable robust comparative analysis of model fairness under both structural and representational interventions.

3.5 Fairness Evaluation with VB-Mitigator

To assess the extent and mitigation of bias in computer vision models, we employ the VB-Mitigator framework as a benchmark fairness evaluation tool. VB-Mitigator provides a standardized suite of fairness metrics and visualization modules for quantifying disparities across protected subgroups. This tool enables both pre- and post-mitigation comparisons, thereby offering insight into how annotation-level imbalances manifest as performance disparities in trained models.

- **Demographic Parity Difference (DPD):** measures differences in the probability of positive classification outcomes across demographic groups, assessing whether each group receives favorable predictions at similar rates.
- **Equal Opportunity Difference (EOD):** measures differences in true positive rates between protected and unprotected groups, highlighting whether one group is systematically advantaged in correct predictions.
- **Average Odds Difference (AOD):** averages the differences in both false-positive and true-positive rates, providing a balanced indicator of predictive fairness.

FACET will be processed through VB-Mitigator both before and after the implementation of fairness interventions such as adversarial debiasing and disparate impact mitigation. By comparing these fairness scores across stages, we can determine whether model-level bias decreases as a result of in-processing debiasing techniques. The analysis will further allow us to examine alignment between annotation-level statistical imbalances and model-level fairness outcomes, strengthening the interpretability of our significance testing and fairness interventions.

4. Preliminary Results

4.1 EDA Findings

The reason we chose to explicitly examine label occurrences is due to how META classified their images and the individuals depicted in them. Mainly, the fact that META chose to use words such as “Astronaut” or “Lawman”. Some categories are prone to societal bias or gendered bias based on wording. We aimed to uncover if META encoded those biases into the data.

From our EDA, we were able to see a significant amount of bias towards masc-presenting individuals, altogether with nearly $\frac{1}{2}$ of all data belonging to masc-presenting individuals. This extends to skin tones as well. Within the Monk skin tone scale, the lowest values in the 1-10 scale represent the lighter skin tones, with number 5 being the midpoint between dark and light. For our purposes, we chose that all values greater than 5 are considered “dark” skin tones. Within the data, the count of individuals with skin pigmentation greater than 5 is out-ranked by a ratio of 3:1, favoring light tones over dark tones. Age labels also raise interpretability concerns. FACET classifies subjects as ‘young,’ ‘middle-aged,’ or ‘old,’ but the dataset documentation does not provide clear criteria for these categories. The predominance of the ‘middle-aged’ label, nearly half the dataset, suggests either imbalanced sampling or broad categorization. Without transparent mapping guidelines, these labels risk obscuring nuanced age-related variation and limiting the validity of fairness analyses involving age.

The EDA revealed that despite META’s preprocessing methods, there still exists bias in the data that must be accounted for. It’s very possible that the bias we discovered was introduced during the pre-processing stage. The hope is that the findings can be mitigated with the strategies mentioned above.

4.2 Significance findings

To statistically validate the visual trends observed in our exploratory data analysis (EDA), we conducted a series of significance tests addressing both representation and contextual bias within the FACET dataset. These analyses quantify the magnitude and statistical certainty of demographic imbalances, revealing the extent to which certain gender, skin tone, and age groups are over or underrepresented and whether these disparities extend into contextual image features such as lighting and visibility.

4.2.1 Representation Bias: Chi-Square Goodness-of-Fit Tests

We first applied Chi-Square Goodness-of-Fit (GOF) tests to determine whether protected attributes were uniformly represented across categories. The null hypothesis assumed equal

distribution among gender, skin tone, and age groups, while the alternative posited significant deviation from uniformity.

Attribute	n	χ^2	p -value	Cohen's w	Interpretation
Gender	43,580	39,705.9	$< 10^{-300}$	0.955	Extremely large disparity
Skin tone	49,536	30,082.2	$< 10^{-300}$	0.779	Very large imbalance
Age	49,182	27,347.9	$< 10^{-300}$	0.746	Very large imbalance

All p -values were effectively zero, indicating that the probability of these imbalances occurring by chance is negligible. Cohen's w effect sizes (0.75–0.95) exceed the “large” threshold, signifying systemic demographic imbalance rather than random fluctuation. These findings confirm that the dataset is not demographically balanced; certain genders, skin tones, and age groups are heavily overrepresented, corroborating the visual disparities observed in the EDA phase.

4.2.2 Category-Level Bias: Pairwise Proportion z-Tests

To identify which specific categories contributed most to the observed imbalances, we performed pairwise proportion z-tests with Holm correction to control for multiple comparisons. These tests revealed that:

- Gender: Masculine-presenting individuals are significantly overrepresented compared to feminine or non-binary groups ($p < 0.001$).
- Skin tone: Lighter tones (categories 1–3) dominate the dataset, while darker tones (8–10) are substantially underrepresented.
- Age: The “young” and “middle-aged” groups occur at much higher rates than “older” groups.

The adjusted p -values (all < 0.05) and large proportional differences confirm that these disparities are not only statistically significant but also practically meaningful, lending statistical rigor to the distributional patterns previously visualized through bar and density plots.

4.2.4 Summary of Statistical Evidence

The convergence of these results provides strong quantitative evidence of both representational and contextual bias within FACET. Table 4.2 summarizes the primary findings:

Bias Type	Evidence	Metrics	Interpretation
Representation Bias	χ^2 GOF tests are significant for all attributes	χ^2, p , Cohen’s w	Unequal representation across demographics
Category-Level Bias	Pairwise z-tests significant	$z, p_{adj}, \Delta\text{proportion}$	Overrepresentation of masculine and lighter skin tone groups
Contextual Bias	χ^2 associations are significant between demographics and visibility/lighting	p , Cramér’s V	Certain groups were photographed in suboptimal conditions

Collectively, these analyses demonstrate that the FACET dataset exhibits statistically and practically significant demographic and contextual imbalances. The magnitude of these effects (Cohen’s $w \geq 0.7$) suggests entrenched structural bias rather than sampling noise. Such biases not only distort representational fairness but also risk exacerbating model-level disparities by encoding inequitable visual conditions into the training process.

5. Preliminary Conclusion

Our analysis of Meta’s FACET dataset reveals strong evidence of both representational and contextual bias. The EDA demonstrated that masculine-presenting individuals, lighter skin tones, and middle-aged subjects are disproportionately represented across the dataset. These imbalances were confirmed through significance testing, where chi-square goodness-of-fit results and large effect sizes (Cohen’s $w = 0.75\text{--}0.95$) indicated severe demographic skew. Pairwise z-tests further established that masculine and lighter-skinned categories dominate the dataset, while chi-square tests of independence revealed contextual disparities. Particularly, those who have darker skin tones are significantly more underrepresented within FACET.

These findings suggest that the FACET dataset not only underrepresents certain demographic groups but also depicts them under qualitatively different visual conditions, introducing multiple layers of bias. Taken together, our findings show that FACET encodes substantial demographic and contextual imbalances that are both statistically significant and practically meaningful. These disparities highlight a gap between FACET’s role as a fairness benchmark and the representational assumptions embedded in its annotations. By integrating dataset-level statistical analysis with model-level evaluation through VB-Mitigator, our work demonstrates how fairness assessments benefit from a lifecycle perspective that examines both data and modeling stages. In ongoing work, we will test whether in-processing debiasing can meaningfully mitigate these

disparities, and will evaluate the extent to which synthetic resampling, adversarial training, and VB-Mitigator’s own mitigation strategies respond differently to underlying annotation biases.

In the next phase of this project, we will apply Adversarial Debiasing and Disparate Impact analysis as in-processing fairness interventions, in combination with the Visual Bias Mitigator (VB-Mitigator) framework. These tools will enable us to evaluate whether algorithmic mitigation can reduce or eliminate the disparities identified in this study while maintaining predictive performance. VB-Mitigator’s fairness metrics will serve as objective benchmarks for pre- and post-mitigation comparison.

By linking dataset-level statistical findings with algorithmic fairness evaluation, our project contributes to the broader understanding of how bias operates through both representation and modeling. The preliminary results demonstrate the importance of rigorous measurement and confirm that fairness must be treated as a lifecycle issue spanning data collection, model design, and evaluation. Our forthcoming analyses will test whether in-processing debiasing techniques can meaningfully mitigate these disparities and provide recommendations for improving fairness in large-scale computer vision datasets like FACET.

6. Updated Project Plan and Timeline

Research Plan

- Week 7: Conduct EDA of annotations (distribution plots, correlations, demographic breakdowns).
- Weeks 8–9: Draft initial analysis of annotation-level bias.
- Week 10: Run FACET through VB-Mitigator to test bias across subgroups.
- Weeks 11–13: Perform deeper analysis on FACET datasets to determine the extent of machine bias.
- Week 14: Compare VB-Mitigator outcomes with annotation-based findings.
- Week 15: Draft results and create visualizations.
- Week 16: Write final report and prepare presentation.

7. Limitations

We anticipate challenges such as the restricted demographic categories available in FACET, potential computational constraints in running VB-Mitigator, and interpretability limits when comparing annotation-level findings to model-level outcomes. These limitations reflect the kinds of lifecycle challenges identified by Suresh and Gutttag, who caution that fairness interventions must be situated within broader systemic contexts [3]. This project also relies inherently on VB-Mitigator as a valid contribution to the project. However, due to human analysis, VB-Mitigator is not only a semi-redundancy but, should it fall in line with our results, an expert tool in assisting fairness research.

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